Fiscal Year:	FY 2013	**Task Last Updated:**	FY 10/10/2012
PI Name:	Olson, Sandra Ph.D.		
Project Title:	Correlation of Normal Gravity Aerospace Materials Flammability Data with Flammability in Microgravity Environments		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Sandra.Olson@nasa.gov	**Fax:**	FY 216 977-7065
PI Organization Type:	NASA CENTER	**Phone:**	216-433-2859
Organization Name:	NASA Glenn Research Center		
PI Address 1:	LTX, Combustion Physics and Reacting Systems Branch		
PI Address 2:	MS 77-5, 21000 Brookpark Rd.		
PI Web Page:			
City:	Cleveland	**State:**	OH
Zip Code:	44135	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2009 Combustion Science NNH09ZTT001N
Start Date:	01/01/2010	**End Date:**	12/31/2014
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Urban, David	**Contact Phone:**	216-433-2835
Contact Email:	david.l.urban@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Extended to 12/31/2014, per PI (Ed., 10/17/13) NOTE: Extended to 12/31/2013, per PI (Ed., 10/10/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Hirsch, David (NTEC, Jacobs Engineering)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NASA STD 6001 Test 1 is the major method used to evaluate flammability of materials intended for use in habitable environments of U.S. spacecraft. The method is an upward flame propagation test initiated in a stagnant environment and using a well-defined igniter flame at the bottom of a vertically mounted sample. A material passes this test if the vertical burn length is less than 15.2 cm (6 inches) and there is no evidence of transfer of burning debris. The test is conducted in the most severe flaming combustion environment (oxygen concentration, pressure) expected in the spacecraft, currently 30% oxygen at 10.2 psia. Recent research has shown that current normal gravity materials flammability tests do not correlate with flammability in ventilated, micro-, or reduced-gravity conditions. The materials selection for spacecraft is based on the assumption of commonality between ground test flammability results and spacecraft environments, which does not appear to be valid. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just "passing" from a flammability point of view. A research program is underway to correlate normal gravity flammability test data with data under ventilated reduced gravity or microgravity conditions. The program will determine materials suitability for microgravity and reduced gravity testing by evaluating oxygen concentration thresholds under various total pressures in normal gravity tests and in spacecraft-specific environments (ventilated microgravity or reduced partial gravity). The normal gravity data will be correlated with flammability data in microgravity environments and the safety factors involved when selecting spacecraft materials based on normal gravity testing will be evaluated. Rigorous correlations between ground test flammability data and data in ventilated spacecraft environments would allow selection of materials with increased fire resistance, and thus decrease the fire risk in space systems. The operational flexibility will also increase since the safety factors will be assessable.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The identification of the fundamental limit in microgravity and its comparison with the normal gravity flammability limit will help us understand flammability in different configurations on Earth. It will improve the safety of astronauts in spaceflight so they can return safely to Earth.
Task Progress:	First and foremost, we seek to develop a fundamental understanding of how NASA’s material flammability test, NASA-STD-6001.A Test 1, relates to the actual flammability of materials in low- and partial-gravity. As part of this goal, we will obtain a better fundamental understanding of material flammability as a function of concurrent flow, oxygen, and pressure. The hypothesis to be tested in this research is that a material’s flammability hazard in microgravity or partial gravity is at least equivalent to and may be greater than its flammability measured in normal gravity. We also hypothesize that most materials exhibit a similar flammability boundary as a function of oxygen and flow. These two hypotheses then imply that there may be a similar magnitude negative oxygen margin of safety for many materials. If true, a common factor of safety is could be defined for materials to ‘de-rate’ them for use in microgravity or partial gravity. To better understand the actual normal gravity flammability limits, a modified Test 1 protocol has been proposed that, as an alternative to qualifying materials as pass/fail in the worst-expected environments, measures the actual upward flammability limit (akin to the Limiting Oxygen Index (LOI)) for the material so that a more accurate assessment of the margin of safety of the material in the real use environment can be made. The method allows the option of selecting better or best materials based on the margin of safety, as opposed to what would be considered just “passing” from a flammability point of view. We have been conducting drop tower tests of materials to evaluate their flammability boundaries in low gravity. We have found that a number of materials have a negative oxygen margin of safety in low gravity. However, drop testing is limited to 5.18 seconds, so only the thinnest of films can be tested, and even then the limits found may be lower than longer duration testing would find, where the flame just takes longer to go out than the drop time. Recently, we have been focusing on defining a flight experiment that to correlate normal gravity flammability test data with data under ventilated microgravity conditions. The flight experiment will determine the minimum in the flammability map for select materials. An oxygen margin of safety will be evaluated. If it is negative, then a safety factor will be defined for the materials to ‘de-rate’ them for use in microgravity or partial gravity.
Bibliography Type:	Description: (Last Updated: 01/12/2026)
Articles in Peer-reviewed Journals	Ferkul PV, Olson SL. "Zero-Gravity centrifuge used for the evaluation of material flammability in lunar-gravity." Journal of Thermophysics and Heat Transfer. 2011 Jul-Sep;25(3):457-61. , Jul-2011
Papers from Meeting Proceedings	Ferkul PV, Olson SL. "Evaluating Materials Flammability in Microgravity and Martian Gravity Compared to NASA Standard Normal Gravity Test." 42nd International Conference on Environmental Systems, San Diego, CA, July 15-19, 2012. Spacecraft Fire Safety forum. 42nd International Conference on Environmental Systems, San Diego, CA, July 15-19, 2012. Paper 2012-3492. , Jul-2012
Papers from Meeting Proceedings	Hirsch D, Juarez F, Olson SL, Harper S, Motto S. "Selected Parametric Effects on Materials Flammability Limits." 41st International Conference on Environmental Systems (ICES), Portland, OR, 17–21 July 2011. Spacecraft Fire Safety forum. 41st International Conference on Environmental Systems (ICES), Portland, OR, 17–21 July 2011. Paper 2011-5067. , Jul-2011

Papers from Meeting Proceedings	Olson SL, Hirsch D. "Geometry Considerations in Evaluating 0g Materials Flammability Limits for Comparison with NASA Upward Flame Proagation (Test 1) Limits." 41st International Conference on Environmental Systems (ICES), Portland, OR, 17–21 July 2011. Spacecraft Fire Safety forum. 41st International Conference on Environmental Systems (ICES), Portland, OR, 17–21 July 2011. Paper 2011-5068. , Jul-2011